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Information Technology Department
COURSE SYLLABUS
Bachelor of Science in Information Technology
First Semester, SY 2025-2026

COLLEGE : Bustos
PROGRAM : Bachelor of Science in Information Technology
COURSE CODE : IT 306
COURSE TITLE : Elective 3 - Programming in Business and Data Analytics
CREDIT UNITS : 1 unit Lab/ 2 units Lecture
PRE-REQUISITE : IT 206 | IT 207 | IT 208 | IT 209
FACULTY : Enger S. Dulay
CONSULTATION HOURS: 3 Hours / Week

COURSE DESCRIPTION:

This course equips students with essential programming skills tailored for business analytics and decision-making in data-driven organizations. It introduces the use of Python and its libraries to collect, process, analyze, and visualize business data. Emphasis is placed on applying control structures, data structures, file handling, and data manipulation to real-world business scenarios. Students will learn how to develop practical scripts for automation, build basic analytical models, and integrate programming with business intelligence tools. Through hands-on coding exercises, case-based activities, and mini-projects, learners will gain the technical proficiency to translate business problems into programmable solutions that support operational efficiency and strategic insight.

Bisyon

Ang Bulacan State University ay isang progresibong institusyong lumilikha ng kaalaman na kinikilala sa daigdig sa pinakamahusay na pagtuturo, nangungunang mga pananaliksik, at maagap na pagtugon sa pangangailangan ng komunidad.

Misyon

Ang Bulacan State University ay may hangaring humubogng mga propesyonal na mahuhusay. etikal, at handang maglingkod, na nakapag-aambag sa patuloy na pag-unladng bayan.

Core Values: ALAB BuISU!

Agap sa Pagbabago
Lingap sa Kapwa at Sangnilikha
Agapayan sa paggawa
Bayanihan sa banyuhay



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The BuISU Ideal Graduates Attributes (BIG A) reflect the graduate's capacity as:

- A. Makadiyos at Makatao (Ethical and Socially Responsible);
- B. Matalino at Malikhain (Critical and Creative Thinker);
- C. Mapanagutan at Maasahan (Accountable and Dependable Professional);
and
- D. Makabansa at Makabuluhan ang Gawa (Service-Oriented and Nationalistic)

Program Educational Objectives (PEO)

Program Educational Objectives (PEO)	University Mission			
	BIG-A	BIG-B	BIG-C	BIG-D
PEO1 Graduates will thrive as industry professionals or technopreneurs by leveraging advanced technical expertise, innovative problem-solving, and strategic thinking to drive technological innovation and business growth in a dynamic global landscape.	X	X	X	X
PEO2 Graduates will uphold integrity and professionalism in their respective fields of specialization, upholding ethical standards, advocating inclusivity, and harnessing technology to drive innovation and meaningful societal impact.	X	X	X	X
PEO3 Graduates will commit to lifelong learning by continuously enhancing their expertise through relevant certifications or microcredentials, advanced education, or active professional engagement, ensuring adaptability to emerging technologies and industry innovations.		X	X	



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Program Outcomes (PO)

On completion of the course, the student is expected to be able to do the following:

Program Outcomes (PO) (Based on your Program CMO)	Program Educational Objectives		
	PEO1	PEO2	PEO3
a. Apply computing, science, and mathematics principles to solve discipline-specific problems.	x		
b. Understand and apply industry standards and best practices effectively.	x		
c. Analyze complex problems, identify computing requirements, and devise appropriate solutions.	x		
d. Identify and analyze user needs to guide the selection, creation, evaluation, and management of computer-based systems.	x	x	
e. Design, implement, and evaluate computer-based systems, processes, or programs to meet specific needs under realistic constraints.	x		
f. Apply current tools, techniques, and practices essential for the IT profession.	x		
g. Function effectively as a member or leader of a development team, recognizing and fulfilling distinct team roles to achieve project goals.		x	x
h. Assist in developing and managing comprehensive IT project plans.	x	x	
i. Communicate effectively with diverse audiences using logical writing, impactful presentations, and clear technical documentation.	x	x	
j. Evaluate the local and global implications of IT on individuals, organizations, and society.		x	
k. Understand and adhere to ethical, legal, security, and social responsibilities in IT practice.		x	
l. Recognize the need for self-directed learning and performance improvement to support ongoing professional development.			x
m. Understand and apply industry standards and best practices effectively.	x	x	
n. Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles		x	x



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Course Outcomes and Relationship to Program Outcomes

Course Outcomes	Program Outcomes													
After completing this course, the student must be able to:	a	b	c	d	e	f	g	h	i	j	k	l	m	n
CO1. Explain the significance of programming in solving business problems and supporting data-driven decision-making.	I		E						I	I				
CO2. Apply core programming concepts—including variables, control structures, functions, and data structures—to automate business workflows.	E		E	I		E								
CO3. Utilize Python libraries and tools to collect, process, analyze, and visualize business data effectively.					E	D								
CO4. Design and implement basic analytics applications involving data input/output, condition-based logic, and business computations.			E	E	D	D		I						
CO5. Develop and evaluate data-driven solutions using programming techniques for real-world business scenarios involving reporting, forecasting, or classification.					D	D	E	E	E			D		

Note: (I) Introductory Course to an Outcome (E) Enabling Course to an Outcome (D) Demonstrative Course to an Outcome

Course Alignment with the United Nations Sustainable Development Goals

United Nations Sustainable Development Goals	
1. No Poverty	☐ Include topics on economic inequality, poverty alleviation, and inclusive economic growth.
2. Zero Hunger	☐ Discuss food security, sustainable agriculture, and nutrition systems.
3. Good Health and Well-Being	☐ Cover health promotion, mental well-being, and universal healthcare systems.
4. Quality Education	☒ Integrate equitable access to education, lifelong learning, and inclusive teaching practices.
5. Gender Equality	☐ Highlight issues of gender equity, women's empowerment, and reducing gender-based discrimination.
6. Clean Water and Sanitation	☐ Address water resource management, sanitation systems, and access to clean water.
7. Affordable and Clean Energy	☐ Explore renewable energy sources, energy



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	efficiency, and sustainable energy solutions.
8. Decent Work and Economic Growth	☒ Focus on employment opportunities, workplace safety, and sustainable economic policies.
9. Industry, Innovation, and Infrastructure	☒ Discuss sustainable infrastructure, industrialization, and innovation.
10. Reduced Inequalities	☒ Analyze policies and practices that address income and opportunity disparities.
11. Sustainable Cities and Communities	☐ Examine urban planning, disaster resilience, and sustainable living solutions.
12. Responsible Consumption and Production	☐ Address waste reduction, sustainable consumption, and ethical production practices.
13. Climate Action	☐ Discuss mitigation and adaptation strategies for climate change.
14. Life Below Water	☐ Explore marine conservation, sustainable fisheries, and ocean health.
15. Life on Land	☐ Cover biodiversity conservation, reforestation, and land-use practices.
16. Peace, Justice, and Strong Institutions	☐ Promote social justice, conflict resolution, and strong governance.
17. Partnerships for the Goals	☐ Include topics on global collaboration, partnership-building, and cross-sectoral initiatives.

Intended Learning Outcomes	Topics	Week	Learning Activities	Suggested Assessment	Plan of Action in Case of Alternative Delivery Mode
1. Explain the importance of the University Vision, Mission, and Core Values and their importance to the program. (CO5) 2. Discriminate between the PEO, PO, and CO of the program and course, respectively, and their relevance in the real world. (CO5)	Orientation • University Vision, Mission, and Core Values • Program PEO, PO & CO • Classroom Policies • Course Contents & Requirements • Grading System • BulSU AI Policy	1	Outcome Alignment Puzzle (PEO, PO, CO Match-Up) Students are grouped and given cards with sets of mixed-up outcome statements. They must sort and justify them into PEO, PO, and CO categories.	Outcome Concept Recitation (5 bonus points) Students respond orally to prompts about PEO, PO, and CO and how they relate to real-world preparation.	Use digital sorting boards (Jamboard, Padlet, Miro). Recitation to be done via MS Teams with chat response options.



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1. Explain the role of programming in business analytics and decision-making. (CO1) 2. Identify sources and types of business data. (CO1)	Intro to Programming for Business and Analytics • Role of programming • Business data types and sources • Analytics workflow overview	2	Discussion with local case examples Data source mapping activity	Reflection paper: How programming supports business functions (20 pts)	Video lesson with Padlet for data source sharing. Submit essay on LMS.
1. Demonstrate the use of variables, data types, and input/output in Python. (CO2)	Python Basics and Input/Output • Variables and data types • Basic operations • Input/output functions	3	Live coding demo Partner coding drills	Basic coding exercise (VAT calculator)	Google Colab pair exercise. Submit .ipynb file to LMS.
1. Apply conditional logic and loops for basic automation. (CO2) 2. Use NumPy for numerical operations. (CO2)	Control Structures, Functions, and NumPy Basics • if-else logic • Loops • Defining functions • NumPy arrays and operations • if-else logic • Loops • Defining functions • NumPy arrays and operations	4	Practice task bank with scaffolded difficulty Mini lecture on NumPy	Function + NumPy activity (25 pts)	Instructor walkthrough videos, shared templates in Google Colab.
1. Apply appropriate data structures to simulate business processes.	Data Structures in Python • Lists, tuples, dictionaries	5	Guided data simulation: Inventory system Code review	Mini app using data structures	Team-based code sharing using shared Colab

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(CO2)	• Nested structures • Dictionary applications in analytics		sessions		notebooks
1. Implement file reading and writing for storage and automation. (CO3)	File Handling in Python • Reading and writing text/CSV files • Using with and handling file paths	6	Edit and improve prebuilt code Build a receipt generator	File processing task	Upload code and sample files to shared folder. Live code check via video call if needed.
1. Perform basic data analysis using Pandas. (CO4)	Data Analysis with Pandas • Data Frames and Series • Importing data • Filtering, sorting, aggregation	7	Dataset lab with Pandas Guided exploration prompts	Data exploration lab	Sample dataset and template notebook provided asynchronously
1. Clean and prepare data for analysis using Pandas. (CO4)	Data Cleaning with Pandas • Handling missing data • Removing duplicates • Data formatting and type conversion	8	Cleaning challenge: messy sales log Lab guide with visualizations	Cleaning a raw dataset (graded)	Notebook check-in and recorded walkthrough before task submission
1. Demonstrate cumulative knowledge through coding tasks and explain capstone goals. (CO5)	Midterm Examination + Capstone Orientation • Covers Weeks 2–8 • Project guidelines and expectations	9	Individual coding midterm Project topic brainstorming	Midterm exam + Project proposal	Written exam via LMS + recorded oral proposal or slide deck submission
1. Create business visualizations using Matplotlib and Seaborn. (CO4)	Data Visualization with Matplotlib and Seaborn • Plot types and	10	Gallery walk: critique sample visuals Visualization tutorial	Visualization project	Upload PNG and code with explanations. Peer comment in

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	customization				LMS forum.
	Integrating visual output with Pandas				
1. Query relational data and integrate with Python. (CO4)	Working with SQL Databases + Python Integration - SQL: SELECT, JOIN, GROUP BY - SQLite + Pandas integration	11	Run queries and export summaries Hands-on exercises with SQLite files	SQL and Python task	Use DB Browser + Google Colab for integrated workflow. Submit full report.
1. Apply time series techniques using Pandas. (CO4)	Time Series Analysis with Pandas - Datetime indexing - Resampling and rolling stats - Simple forecasts - Recall: BI Tools Summary	12	Calendar dataset walkthrough Time series error correction	Mini forecasting project	Instructor provides template; students edit and submit improved version.
1. Build classification models using scikit-learn. (CO4)	Supervised ML: Classification - KNN classifier - Confusion matrix - Data split and accuracy - Recall: Statistics Summary	13	Live model-building demo Mini project with labeled dataset	ML task: Classify outcomes	Submit confusion matrix screenshot + brief explanation
1. Apply regression models to predict numeric values. (CO4)	Supervised ML: Regression - Linear regression - Model evaluation metrics (MSE, RMSE)	14	Regression prediction scenario Peer validation of results	ML task: Predict numeric values	Use LMS quiz form to upload predictions + interpretation text box.
1. Apply unsupervised learning for business segmentation. (CO4)	Unsupervised ML: Clustering & PCA K-Means clustering	15	Clustering e-commerce dataset Visualize and interpret	Clustering lab	Students submit cluster plot + 2 insights in reflection

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	• PCA overview • Visualization of clusters		clusters		form.
1. Explain and experiment with deep learning for simple business problems. (CO4)	Intro to Deep Learning with Keras • Neural network basics • Activation functions • Simple Keras model training & evaluation	16	Live demo of binary classification using Keras Debugging activity	DL mini-lab	Use Google Colab. Students fork template and adjust one hyperparameter.
1. Present and defend an original data analytics solution. (CO5)	Project Presentation • Project demo and defense • Feedback and peer evaluation	17	Project presentation with Q&A Peer critique using rubric	Final project presentation	Present live or submit recording + documentation for asynchronous review.
1. Demonstrate holistic understanding of analytics programming. (CO5)	Final Examination • Full-course assessment (Python + Data Analysis + ML + DL)	18	Written exam + practical task Post-exam reflection	Final written + practical exam	Timed online test + coding notebook. Follow-up Q&A for verification.

FINAL COURSE OUTPUT: Data-Driven Business Analytics Solution

Rationale:

Modern organizations rely on data-driven strategies to maintain competitiveness and improve operational efficiency. This project integrates students' knowledge and skills in Python programming, data processing, and analytics to simulate real-world problem-solving in a business context.

Students will extract, clean, analyze, and visualize business data using Python and its associated libraries such as Pandas, Matplotlib, Seaborn, and scikit-learn. The output should demonstrate proficiency in working with structured datasets and applying appropriate techniques, such as data cleaning, descriptive analysis, classification or regression, and dashboard creation.



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This capstone output serves as a culminating requirement to validate the student's ability to convert raw data into actionable insights using programming—a critical skill for modern business analysts.

Learning Objectives:

1. Perform end-to-end data processing using Python.
2. Apply descriptive and predictive analytics techniques to a chosen business dataset.
3. Develop an interactive dashboard that presents insights from cleaned data.
4. Demonstrate effective communication and documentation of analytics findings.

Instructions:

Students may work **individually or in pairs**, depending on class size and instructor discretion.

Select a **realistic or actual business dataset**. Open datasets may be sourced from the following:

- Kaggle Datasets
- Data.gov
- UCI Machine Learning Repository

Approval of dataset is required before proceeding with analysis.

The final output must include:

- **Cleaned Dataset (.csv or .xlsx)**
- **Python scripts (.ipynb or .py)**
- **Dashboard (image and/or .html via Plotly or exported visualizations)**
- **Written documentation (PDF)**
- **Video Presentation (max 5 minutes)**



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SCORING RUBRICS FOR THE ANALYTICS OUTPUT

CRITERIA	WEIGHT
Dataset volume and business relevance	10
Quality of data cleaning and preprocessing (missing data, formats, outliers)	15
Correct application of descriptive statistics and business metrics	25
Relevance and clarity of visualizations	20
Proper use and explanation of ML model (classification/regression)	20
Clarity of insights and recommendations for business improvement	10
Technical documentation (code comments, logical flow, report)	50
TOTAL	150

SCORING RUBRICS FOR THE PROJECT PRESENTATION

CRITERIA	WEIGHT
Video explanation of data processing and ML technique	30
Presentation and discussion of final dashboard and findings	30
Q&A performance during final defense (clarity, accuracy, confidence)	20
Individual contribution and presence during the project (peer/self/instructor evaluation)	20
TOTAL	100

OTHER REQUIREMENTS AND ASSESSMENTS:

Aside from the final output, the student will be assessed at other times during the term by the following:

- Active class participation/Work ethics
- Activities
- Major Examinations (Midterm and Final Term)
- Small Group Discussion
- Quizzes
- Exit Assessment

GRADING SYSTEM:

Major Examinations (Midterm and Final Exam)	-	30%
Laboratory Activities / Project	-	40%
Quizzes / Proficiency Tests	-	20%
Attendance/Recitation / Class Participation	-	10%
Total	-	100%



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GRADE EQUIVALENT:

Range	Grade
96.50-100.00	1.00
93.50-96.49	1.25
90.50-93.49	1.50
87.50-90.49	1.75
84.50-87.49	2.00
81.50-84.49	2.25
78.50-81.49	2.50
75.50-78.49	2.75
75.00-75.49	3.00
Conditional Passed	4.00
74.00 and below	5.00

REFERENCES:

1. McKinney, W. (2022). Python for Data Analysis: Data Wrangling with Pandas, NumPy, and Jupyter. 3rd ed. O'Reilly Media.
2. Grus, J. (2019). Data Science from Scratch: First Principles with Python. 2nd ed. O'Reilly Media.
3. VanderPlas, J. (2016). Python Data Science Handbook. O'Reilly Media.
4. Provost, F., & Fawcett, T. (2013). Data Science for Business: What You Need to Know about Data Mining and Data-Analytic Thinking. O'Reilly Media.
5. James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). An Introduction to Statistical Learning with Applications in R. Springer.
6. **Galang, G. (2025). *Programming in Business Analytics*. Bulacan State University.

Online Resources:

1. W3Schools – Python Tutorial
2. Kaggle Learn – Python, Pandas, and Machine Learning Courses
3. DataCamp – Intro to Python, Pandas, and Data Visualization
4. Coursera – Applied Data Science with Python Specialization

Datasets for Project and Practice:

1. Kaggle Datasets
2. OpenML Data Repository
3. Data.gov.ph – Philippine Government Open Data

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4. Google Dataset Search
5. [Tableau Public Data Sets](#)

CLASS POLICIES:

1. The University rules and regulations should be observed at all times.
2. Students should register for the online class created by the Professor on or before the first day of classes.
3. The Professor will check the attendance at every meeting. Accumulating 20% of absences will automatically result in the student being dropped from the class.
4. Students are expected to come to class ready at all times. Active participation will be encouraged in all class discussions.
5. Students should submit all requirements on time.
6. Any form of academic dishonesty, including plagiarism, is not tolerated.
7. For take-home assessments, students may be required to submit plagiarism and AI detection results, as well as the extent of AI usage based on the frameworks outlined in the BulSU AI Policy.

Prepared by:

DIGNA S. EVALE, DIT
ENGER S. DULAY

Faculty

Evaluated by:

LYN M. DALISAYMO, MIT

Program Chair, BSIT

Approved by:

MARICEL E. SIERTO, Ed. D.

Dean, Bustos Campus



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Declaration

I have read and understood the above syllabus in full and in participating in this course I agree to the above rules. I have a clear understanding of the policies and my responsibilities, and I have discussed everything unclear to me with the professor.

I will adhere to the academic integrity and policy and I will treat my fellow students and my professor with due respect.

I understand that this syllabus can be modified or overruled by announcements of the professor in class or on any social media site at any time

Student's Printed name Signature Date

Student's Copy

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Declaration

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